HAEBOT / KIRO V2 power system comparison

Engineering review document, Rev B (Anker design Rev B, EV design Rev A), 2026-11-13

Both designs are PROPOSED. Neither has been built or measured. KiCad ERC has not been run yet.

1. Project overview

- HAEBOT (also called KIRO) is a mobile dual-arm robot: Jetson Orin Nano Super, SO-101 arms, four mecanum wheels driven by ST3215-C018 bus servos through one Bus Servo Adapter board.
- V1 runs from AC mains through a 12 V SMPS, two fuse boxes and a 12 V to 5 V buck converter. It is built and is the reference for all load figures.
- V2 removes the mains cable. Two batteries are under consideration, so there are two independent designs.
- **V2-A** uses an Anker SOLIX C300 DC (A1726) power station with ready-made leads on its protected ports.
- **V2-B** uses a 12 V 45 Ah battery, keeps the V1 fuse-box layout and moves both arms to 12 V servos.

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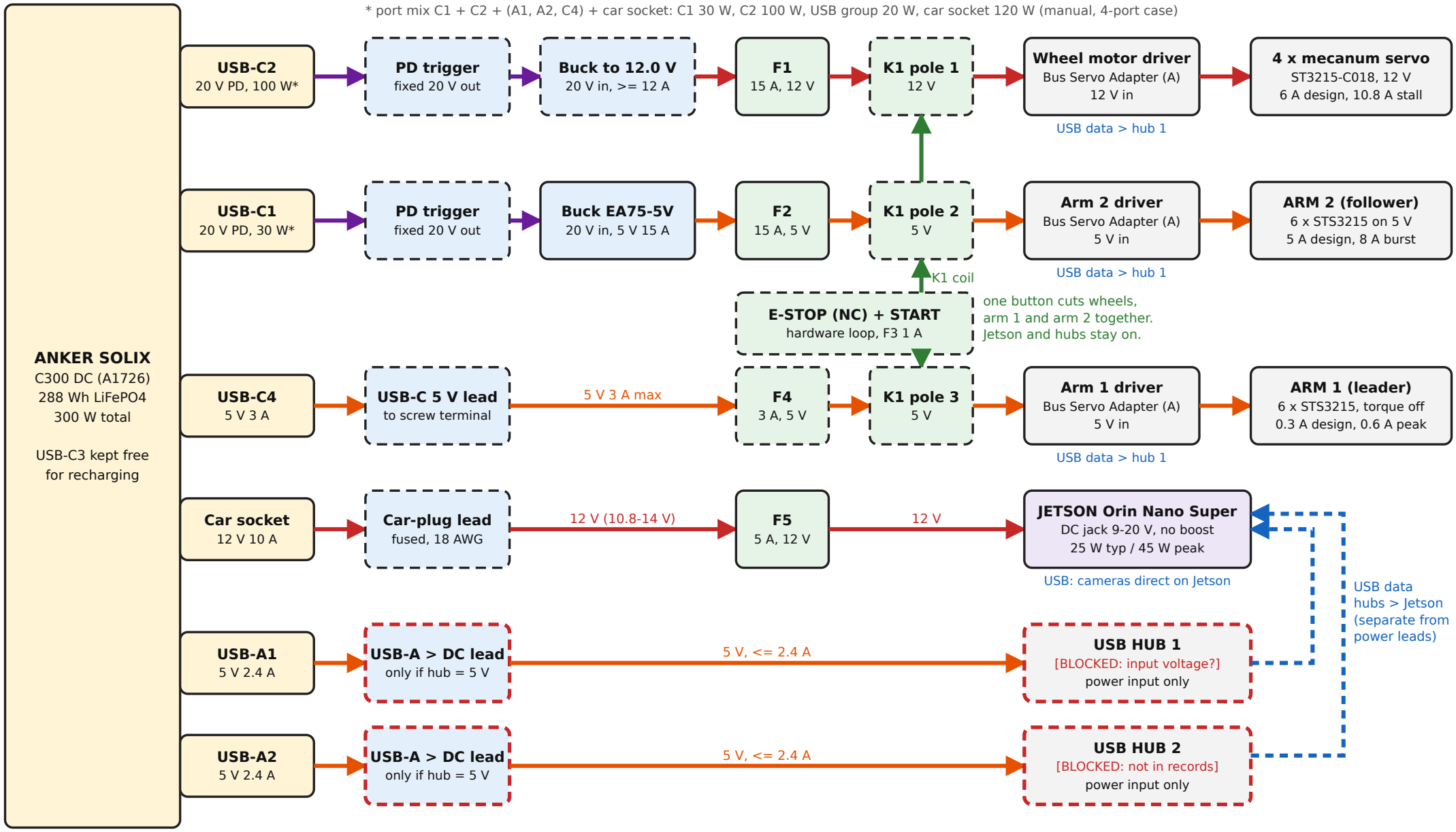
- Page 2: Anker block diagram. Page 3: Anker electrical schematic (A3).
- Page 4: EV battery block diagram. Page 5: EV battery electrical schematic (A3).
- Then: architecture comparison, power budget comparison, component and connection summary, electrical warnings, unresolved hardware items.

How to read the status words

Word	Meaning
Verified	Supported by a datasheet, a manual or a measurement recorded in the repository
Calculated	Derived from verified values with stated assumptions
Proposed	Not bought, not built, not tested
Blocked	Cannot be finalised until a hardware fact is supplied

HAEBOT / KIRO V2-A Anker SOLIX C300 DC power block diagram

Rev B, 2026-11-13. PROPOSED architecture, not bench-verified. Port limits from the Anker A1726 user manual; loads from V1 Rev F records.



LEGEND — 20 V USB-C PD — 12 V — 5 V — USB data — protection / E-stop control

Borders: solid = existing V1 hardware dashed = PROPOSED (not bought / tested) red dashed = BLOCKED (needs a hardware fact)

Not supported as requested: 12 V PD trigger (station has no 12 V USB-C profile) and 5 V PD trigger for a follower arm (5 V PD is 3 A max).

HAEBOT / KIRO V2-A ANKER SOLIX C300 DC POWER SYSTEM

Electrical schematic, Rev B, 2026-11-13. STATUS: PROPOSED - build only after the bench checks in docs/verification_report.md

BRANCH D - WHEEL DRIVE: USB-C2 -> 20 V PD trigger -> 12 V buck -> F1 -> K1 (E-stop) -> wheel adapter -> 4 x ST3215-C018



BRANCH C - ARM 2 (powered follower): USB-C1 -> 20 V PD trigger -> EA75-5V buck -> F2 -> K1 (E-stop) -> arm adapter

E-STOP CONTROL (hardware only, no software): DRIVE_12V -> F3 -> S1 (NC) -> START || K1 aux -> K1 coil. Opens wheels, arm 1 and arm 2. Jetson + hubs stay on.

BRANCH B - ARM 1 (leader / torque-off arm only): USB-C4 5 V 3 A -> USB-C 5 V lead -> F4 -> K1 (E-stop) -> arm adapter

BRANCH E - JETSON: car socket 12 V 10 A -> fused plug lead -> F5 -> DC jack 5.5 x 2.5 mm centre + (devkit jack accepts 9-20 V; no boost)

BRANCH A - USB HUBS: USB-A1 / USB-A2 (5 V 2.4 A each) -> DC lead -> hub power input. BLOCKED: hub input voltage unknown

ANKER PORT LIMITS (manual A1726)

USB-C1: 5/9/15 V 3 A, 20 V 5 A (100 W)
USB-C2/C3: + 28 V 5 A (140 W)
USB-C4: 5 V 3 A (15 W)
USB-A1/A2: 5 V 2.4 A (12 W each)
USB-A1 + A2 + C4 together: 20 W
Car socket: 12 V 10 A (120 W)
Total output: 300 W
NO 12 V USB-C profile, no PPS listed

PORT MIX USED ON THIS SHEET

C1 + C2 + (A1,A2,C4) + car socket
= manual 4-port case:
C1 30 W | C2 100 W | USB group 20 W
| car 120 W

CALCULATED MARGINS

Wheels: 78 W design < 100 W OK
Wheels: 141 W all-stall > 100 W
Arm 2: 28 W design < 30 W marginal
Arm 2: 44 W burst > 30 W
Jetson: 45 W peak < 120 W OK
Hubs + Arm 1: must stay < 20 W

WARNINGS

1. Never join the + of two ports.
2. Plugging or unplugging any port can renegotiate PD and drop the other outputs for a moment.
3. Set and measure U3 (12.0 V) and U4 (5.0-5.2 V) before any servo is connected.
4. Negatives of all ports become common through USB shields. Motor current must return on its own pair, not through USB.
5. E-stop cuts wheels + both arms.

STATUS WORDS

EXISTING = part on the V1 robot
PROPOSED = not bought, not tested
BLOCKED = needs a hardware fact

Drawn 2026-11-13. KiCad ERC must be run in KiCad 8+ before release
Single lines to servos = 3-wire Molex 5264 bus cable (GND / VCC / SIG)
Port limits: Anker A1726 user manual. Loads: V1 Rev F records (kiro-wiring/docs)
Status: PROPOSED - not bench-verified. See docs/verification_report.md
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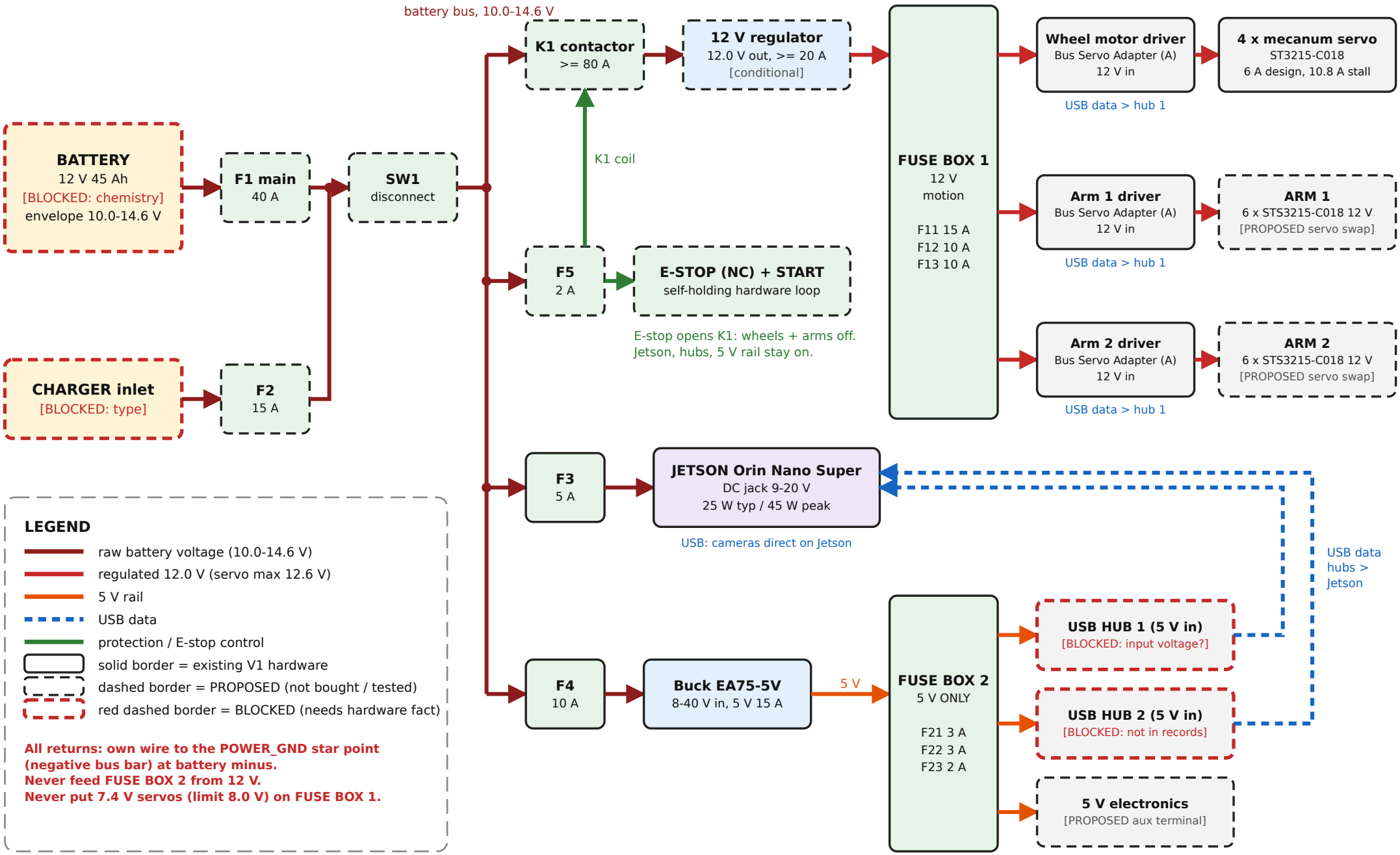
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File: HAEBOT_ANKER.kicad_sch

Title: HAEBOT / KIRO V2-A - Anker SOLIX C300 DC power system

Size: A3	Date: 2026-11-13	**Rev: B**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1

HAEBOT / KIRO V2-B 12 V 45 Ah EV battery power block diagram

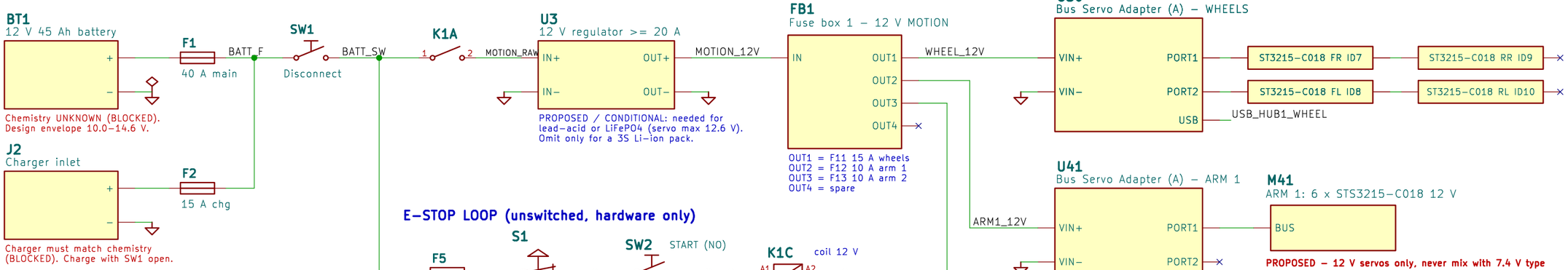
Rev A, 2026-11-13. PROPOSED architecture. Battery chemistry, BMS limits and charger are BLOCKED items; fuse and wire values are preliminary.



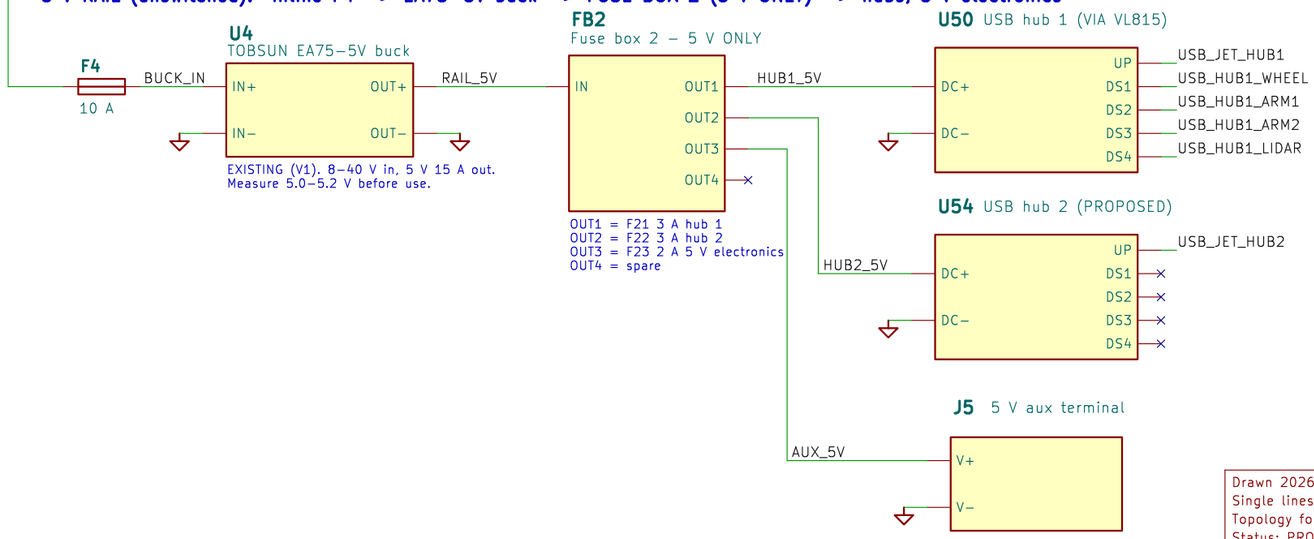
HAEBOT / KIRO V2-B 12 V 45 Ah EV BATTERY POWER SYSTEM

Electrical schematic, Rev A, 2026-11-13. STATUS: PROPOSED - battery chemistry, BMS limits and charger are BLOCKED items

MOTION POWER (cut by E-stop): battery -> F1 -> SW1 -> K1 -> 12 V regulator -> FUSE BOX 1 -> wheels, arm 1, arm 2



5 V RAIL (unswitched): inline F4 -> EA75-5V buck -> FUSE BOX 2 (5 V ONLY) -> hubs, 5 V electronics



SYMBOLS AND STATUS

Arrow-down symbol = POWER_GND:
own return wire to the star point
(negative bus bar) at battery -.
Never return motor current
through a USB cable.

EXISTING = on the V1 robot
PROPOSED = not bought / tested
BLOCKED = needs a hardware fact

E-STOP BEHAVIOUR

S1 pressed or wire broken:
K1 opens, wheels + both arms
lose power (arms drop).
Jetson, hubs, 5 V stay on.
Release S1: no restart until
START is pressed.

BEFORE BUILDING

Read battery + charger labels.
Fix F1, F2, K1, U3 after that.

On hub 1, port DS4:
USB_HUB1_LIDAR -> RPLIDAR C1 + CP2102N

DESIGN VALUES

Jetson 12 V: 25 / 36 / 45 W
Wheels: 1.2 / 6.0 / 10.8 A
Arm (12 V, each): 3 A design,
8 A peak (ESTIMATE)
5 V rail: about 10 W (assumed)
Battery: see docs/power_budget.md

WIRE (from V1 Rev F rules)

Battery to FB1: 8 AWG
Wheels, arms, buck: 14 AWG
Jetson: 18 AWG
Control loop: 20 AWG

WARNINGS

1. Never feed FB2 from 12 V.
2. Never connect 7.4 V servos
(limit 8.0 V) to FB1.
3. ST3215 max 12.6 V: battery
may reach 14.6 V -> U3.

Drawn 2026-11-13. KiCad ERC must be run in KiCad 8+ before release
Single lines to servos = 3-wire Molex 5264 bus cable (GND / VCC / SIG)
Topology follows V1 Rev F (two fuse boxes, EA75-5V buck); SMPS replaced by battery
Status: PROPOSED - battery chemistry, BMS and charger are not verified (BLOCKED items)
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File: HAEBOT_EV.kicad_sch

Title: HAEBOT / KIRO V2-B - 12 V 45 Ah EV battery power system

Size: A3	Date: 2026-11-13	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1

2. Architecture comparison

	V2-A Anker SOLIX C300 DC	V2-B 12 V 45 Ah EV battery
Energy source	288 Wh LiFePO4 station, 300 W total DC (verified, manual)	12 V 45 Ah, 540-576 Wh nominal; chemistry not identified (blocked)
Distribution	five station ports, one lead pair per branch	main fuse, disconnect, battery bus, two fuse boxes
Wheel supply	USB-C2 20 V PD, PD trigger, 12.0 V buck, F1 15 A	K1, 12.0 V regulator (conditional), fuse box 1 F11 15 A
Arm supply	existing 5 V servos. Leader on USB-C4 (5 V 3 A). Follower on USB-C1 through 20 V trigger and EA75-5V buck	new 12 V servos STS3215-C018 on fuse box 1, F12 and F13 10 A
Jetson supply	car socket 12 V 10 A, direct to the 9-20 V DC jack, F5 5 A	battery bus, direct to the 9-20 V DC jack, F3 5 A
5 V rail	EA75-5V for the follower arm only	EA75-5V, fuse box 2: hubs and 5 V electronics
USB hubs	USB-A ports (blocked: hub input voltage unknown)	fuse box 2 (blocked: hub input voltage unknown)
Emergency stop	3-pole contactor opens wheels, arm 1 and arm 2 together; hardware loop	battery contactor opens fuse box 1; hardware loop
Over-current protection	station port protection plus branch fuses	40 A main fuse plus branch fuses (values preliminary)
Charging	USB-C2 / USB-C3, 140 W PD each	blocked: charger not identified
New parts to buy	2 PD triggers, 12 V buck, contactor, E-stop, START, leads, holders	main fuse, disconnect, contactor, regulator, bus bar, E-stop, START, 12 servos
Changes to the robot	none to arms or software	12 servos replaced, both arms recalibrated
Main limit	port sharing: 100 W for wheels and 30 W for the follower in this port mix	battery data missing; servo rail must stay at or below 12.6 V

3. Power budget comparison

Loads are the V1 Rev F values. Anker usable energy: 288 Wh x 0.85 = 245 Wh (assumption). EV usable energy: Lead-acid (AGM / gel) 270 Wh, LiFePO4 4S 461 Wh, Li-ion 3S 400 Wh (assumed usable shares 50 / 80 / 80 %).

Scenario	Anker: power	Anker: runtime	EV: power	EV: Lead-acid	EV: LiFePO4 4S	EV: Li-ion 3S
1. Idle robot	38 W	6.4 h	36 W	7.5 h	12.8 h	11.1 h
2. Jetson and electronics active	50 W	4.9 h	64 W	4.2 h	7.3 h	6.3 h
3. Two arms operating	81 W	3.0 h	134 W	2.0 h	3.4 h	3.0 h
4. Robot driving, arms holding	137 W	1.8 h	147 W	1.8 h	3.1 h	2.7 h
5. Typical combined operation	63 W	3.9 h	73 W	3.7 h	6.3 h	5.4 h
6. High load	157 W	1.6 h	281 W	1.0 h	1.6 h	1.4 h

- The scenario definitions differ slightly between the designs (the EV design has two powered 12 V arms). Details are in each folder's docs/power_budget.md.
- Anker wheel branch: 78 W at design load against a 100 W port limit; 141 W with all four wheels stalled is not supported.
- Anker follower branch: 28 W at design load against 30 W; the 44 W burst is not supported.
- EV battery current: 23 A in the high-load case; 34 A at 12.0 V if everything peaks together.

4. Component and connection summary

Function	V2-A Anker	V2-B EV battery	Connection
Computer	Jetson Orin Nano Super Dev Kit (existing)	same	5.5 x 2.5 mm barrel, centre +
Wheel motor driver	Bus Servo Adapter (A), CH343 (existing, one board)	same	screw terminal, USB-C data
Wheel motors	4 x ST3215-C018 (existing)	same	Molex 5264, two chains
Arm drivers	2 x Bus Servo Adapter (A) on 5 V (existing)	same boards moved to 12 V	screw terminal, USB-C data
Arm servos	STS3215 7.4 V type on 5 V (existing)	12 x STS3215-C018 12 V (proposed)	Molex 5264
5 V converter	TOBSUN EA75-5V, fed with 20 V (existing)	TOBSUN EA75-5V, fed from the battery (existing)	flying leads
12 V regulation	buck 20 V to 12.0 V, 12 A or more (proposed)	buck-boost 12.0 V, 20 A or more (proposed, conditional)	screw terminals
Fuse boxes	inline holders only	2 x Dorhea 4-way (existing)	blade fuses
E-stop	mushroom NC, START, 3-pole contactor (proposed)	mushroom NC, START, battery contactor (proposed)	screw terminals
USB hubs	hub 1 VIA VL815 (existing), hub 2 (proposed)	same	data to Jetson; power input blocked

5. Important electrical warnings

- Follower servos of the 7.4 V type trip above 8.0 V. Never connect them to a 12 V rail.
- ST3215-C018 servos accept at most 12.6 V. A lead-acid or LiFePO4 battery exceeds this; use the 12.0 V regulator.
- The Anker station has no 12 V USB-C profile. A 12 V PD trigger will not give 12 V.
- USB-C delivers at most 3 A at 5 V. A powered arm cannot run from a 5 V USB-C source.
- Never join the positive outputs of two station ports. Never feed fuse box 2 from 12 V.
- Motor current must return on its own wire, never through a USB cable.
- The E-stop is a hardware loop. When it opens, the powered arms drop under gravity.
- A 45 Ah battery can deliver hundreds of amperes into a short. Fit the main fuse within 150 mm of the positive terminal.

6. Unresolved hardware verification items

#	Item	Affects	What settles it
1	EV battery chemistry, voltage range, BMS limits, connector	V2-B: main fuse, regulator, cut-off	photo of the battery label
2	Charger type and connector	V2-B: charging connection, F2	photo of the charger label
3	USB hub input voltage, current, plug	both: hub power branch	photo of hub and adapter labels
4	Second USB hub: does it exist?	both	your answer
5	Which arms are arm 1 and arm 2 (leader / follower)	V2-A port assignment; V2-B servo count	your answer
6	Trial fit and current of one STS3215-C018 in an arm joint	V2-B: brackets, F12, F13	bench test
7	20 V available on USB-C1 when capped at 30 W; behaviour on hot-plug	V2-A: follower branch	bench test with a USB-C meter
8	Station port negatives common internally?	V2-A: grounding	continuity measurement
9	Part numbers: PD triggers, bucks, contactors, E-stop, disconnect	both BOMs	selection and purchase
10	Jetson carrier is the stock P3768 board	both: 9-20 V jack range	visual check
11	KiCad ERC in KiCad 8 or newer	both schematics	run ERC, review every message
12	Recommended Anker alternative: arm buck on the car socket	V2-A: 140 W for the wheels	your decision